

PYTHON INPUT AND OUTPUT COMPLETE NOTES

What is Input in Python?

Input means taking data from the user during program execution. Python provides the input() function to receive data from the keyboard.

Syntax:

```
variable = input("message")
```

Example:

```
name = input("Enter your name: ")  
print(name)
```

Output:

Enter your name: Gnanesh

Gnanesh

Important Point:

The input() function always returns data in string format.

Example:

```
a = input("Enter first number: ")  
b = input("Enter second number: ")
```

```
print(a + b)
```

Output:

Enter first number: 10

Enter second number: 20

1020

Explanation:

Python treats 10 and 20 as strings and concatenates them.

Type Conversion:

Type conversion changes one data type into another.

Integer Conversion Using int():

```
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
print(a + b)
```

Output:

Enter first number: 10

Enter second number: 20

30

Float Input:

```
salary = float(input("Enter salary: "))  
print(salary)
```

Output:

Enter salary: 55000.75

55000.75

String Input:

```
city = input("Enter city: ")  
print(city)
```

Output:

Enter city: Hyderabad

Hyderabad

Multiple Inputs Using split():

```
a, b = input("Enter two values: ").split()  
print(a)  
print(b)
```

Output:

Enter two values: 10 20

10

20

Multiple Integer Inputs Using map():

```
a, b = map(int, input("Enter two numbers: ").split())
```

```
print(a + b)
```

Output:

Enter two numbers: 10 20

30

ValueError:

Occurs when invalid data is converted.

```
age = int(input("Enter age: "))
```

Input:

abc

Error:

ValueError

Solution:

try:

```
    age = int(input("Enter age: "))
```

```
    print(age)
```

except ValueError:

```
    print("Enter valid number")
```

Output:

Enter age: abc

Enter valid number

EOFError:

Occurs when no input is provided.

try:

```
    name = input()
```

except EOFError:

```
    print("No input provided")
```

What is Output in Python?

Output means displaying information to the user. Python uses the `print()` function for output.

Syntax:

```
print(value)
```

Simple Output Example:

```
print("Hello Python")
```

Output:

Hello Python

Printing Variables:

```
name = "Gnanesh"
```

```
age = 23
```

```
print(name)
```

```
print(age)
```

Output:

Gnanesh

23

Printing Multiple Values:

```
print("Python", 100, 55.5)
```

Output:

Python 100 55.5

sep Parameter:

The sep parameter separates multiple values.

Example:

```
print(10, 20, 30, sep="-")
```

Output:

10-20-30

end Parameter:

The end parameter controls the ending character.

Example:

```
print("Hello", end=" ")
```

```
print("Python")
```

Output:

Hello Python

Escape Characters:

\n -> New line

\t -> Tab space

\\ -> Backslash

\' -> Single quote

\\" -> Double quote

New Line Example:

```
print("Hello\nPython")
```

Output:

Hello

Python

Tab Example:

```
print("Hello\tPython")
```

Output:

Hello Python

Formatted Output Using f-string:

```
name = "Gnanesh"
```

```
age = 23
```

```
print(f"My name is {name} and age is {age}")
```

Output:

My name is Gnanesh and age is 23

NameError:

Occurs when quotes are missing.

Wrong Example:

```
print(Hello)
```

Error:

NameError

Correct Example:

```
print("Hello")
```

Output:

Hello

Difference Between Input and Output:

Input:

- Takes data from user
- Uses input()
- Receives information

Output:

- Displays data to user
- Uses print()
- Shows information